

DEVELOPING E-LEARNING CONTENT FOR GOVERNMENT SCHOOLS

A Community Engagement Project Report

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Abstract

The Shiksha Setu project is a comprehensive, community-driven e-learning platform specifically developed to address the acute digital education gap prevalent in government schools across Maharashtra, India. This project was conceived and executed as part of the Community Engagement Project (CEP) curriculum of the B.Sc. Information Technology programme at S.K. College of Science & Commerce, Navi Mumbai, affiliated to the University of Mumbai.

Government schools, which cater to the majority of students from economically weaker sections of society, often lack access to quality digital learning resources. Teachers at these institutions face challenges in creating structured, multimedia-enriched lesson content due to time constraints and limited technical expertise. Students, in turn, miss out on the engaging, interactive learning experiences that their counterparts in private schools routinely access.

Shiksha Setu — meaning 'Education Bridge' in Hindi — directly addresses these gaps by providing a dual-interface platform: a Teacher Portal that enables educators to author, organize, and assign structured digital lessons without any programming knowledge, and a Student Portal that gives learners in Classes 6 through 10 access to subject-specific lessons, quizzes, digital library resources, achievement tracking, and several innovative learning modules designed to enhance engagement and comprehension.

The platform is built entirely with web technologies — HTML5, CSS3, and JavaScript — making it deployable without costly server infrastructure and accessible even on basic smartphones and low-bandwidth connections. The backend is powered by Node.js and PostgreSQL, providing robust data management, offline synchronization through IndexedDB, and multi-language support in English, Hindi, and Marathi.

Version 2.0 of the platform introduces five major new interactive modules: an AI Doubt Solver for instant question resolution, a Class Leaderboard for motivation and healthy academic competition, a Virtual Science Lab for hands-on simulation-based experiments, a Study Planner for time management and goal-setting, and a Vocabulary Builder with flashcard-based learning for English and subject-specific terminology mastery.

This report documents the complete design, architecture, implementation methodology, new feature additions, and deployment strategy for the Shiksha Setu platform, providing a detailed account of how technology was leveraged to create a meaningful community impact.

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Chapter 1: Introduction

1.1 Background and Context

India is home to over 1.5 million government schools that collectively serve nearly 250 million students from economically disadvantaged and rural backgrounds. Despite government initiatives such as the Digital India programme, the Samagra Shiksha Abhiyan, and the National Education Policy 2020, a significant proportion of government school students continue to receive education in largely text-heavy, paper-based environments. The contrast with the digitally enriched educational experiences available in private institutions is stark, creating what educationists refer to as the 'digital education divide'.

The Shiksha Setu project was born out of a direct observation of this divide. During community fieldwork conducted by the project team at government schools in the Navi Mumbai region, it became evident that while mobile smartphones were increasingly common even among lower-income households, structured digital learning content tailored to the government school curriculum and delivered in local languages was virtually absent. Students had access to devices but lacked meaningful educational applications that aligned with their syllabus and were intuitive enough for self-directed learning.

The project team, comprising B.Sc. IT students with backgrounds in web development, database management, and user interface design, recognized a unique opportunity to apply their technical skills for direct community benefit. The result is Shiksha Setu — a bilingual, offline-capable, curriculum-aligned e-learning platform that bridges the gap between available technology and quality educational content for government school students in Classes 6 through 10.

1.2 Project Scope and Coverage

Shiksha Setu is designed to serve two primary user groups: teachers who need a simple content authoring and management system, and students who need an engaging, self-paced learning environment. The platform covers six core subjects — Mathematics, Science, English, Hindi, Social Studies, and Computer Science — across five grade levels, from Class 6 to Class 10. This spans the complete secondary education curriculum as prescribed by the Maharashtra State Board of Secondary and Higher Secondary Education (MSBSHSE).

Beyond curriculum coverage, the platform extends its scope through specialized interactive modules that address specific learning challenges. The AI Doubt Solver provides immediate question-answering support for students who cannot readily access teacher assistance at home.

The Virtual Science Lab offers hands-on experimentation for students whose schools lack laboratory infrastructure. The Study Planner encourages disciplined, self-regulated learning habits. The Vocabulary Builder addresses the persistent challenge of building academic language proficiency. And the Leaderboard fosters a spirit of healthy academic competition that motivates consistent engagement.

The administrative scope of the platform includes school-level management features — assigning courses to classes, monitoring student progress at an aggregate level, and generating completion certificates — that empower teachers and school administrators to leverage data for pedagogical improvement.

1.3 Report Organization

This report is organized into seventeen chapters. Following this introduction, Chapter 2 presents the detailed problem statement and community need analysis. Chapters 3 and 4 define the project objectives and review relevant prior art and literature. Chapters 5 through 12 detail the design, implementation, and functionality of each major system module including all five new interactive learning modules introduced in Version 2.0. Chapters 13 through 15 describe the technical architecture, database design, offline capabilities, and accessibility features. Chapter 16 documents the testing approach and deployment outcomes, and Chapter 17 concludes the report with reflections and future development directions.

Chapter 2: Problem Statement & Community Need Analysis

2.1 The Digital Education Divide in Government Schools

The fundamental problem addressed by this project is the inequitable distribution of quality digital educational resources between government and private schools in urban India. Private schools routinely deploy interactive smart boards, subscribe to premium e-learning platforms, and provide students with structured digital homework and revision tools. Government schools, by contrast, continue to rely primarily on printed textbooks and blackboard instruction, with little to no structured digital supplementation.

This inequality manifests most acutely in the area of student self-study. Private school students can access subject-specific video explanations, practice questions, and progress tracking tools from home. Government school students typically cannot. The consequence, particularly for the growing number of first-generation learners whose parents cannot provide academic guidance at home, is a significant and measurable gap in academic outcomes.

A needs assessment survey conducted by the project team across three government schools in Navi Mumbai revealed that 78 percent of students owned or had access to a smartphone, but 91 percent were unaware of any government school-aligned digital learning resource available in their regional language. Teachers at these schools reported spending significant time manually creating test papers and revision notes, with no systematic way to share these resources digitally with students.

2.2 Teacher Content Creation Challenges

Government school teachers face a unique set of challenges that distinguish their situation from teachers in well-resourced private institutions. Class sizes are routinely large — often exceeding forty students per teacher — reducing the time available for individualized instruction. Administrative burdens including attendance management, record keeping, and examination preparation consume significant portions of a teacher's working day.

Against this backdrop, the expectation that teachers should also create high-quality digital learning content represents an unrealistic demand without appropriate tooling. Existing content management systems available to educators are predominantly designed for technically proficient users and assume a level of digital literacy that many government school teachers, particularly those in their mid-career or later years, do not possess. There was a clear and demonstrated need for a content authoring interface that is simple enough for a teacher with

basic smartphone familiarity to navigate, yet powerful enough to produce structured, curriculum-aligned digital lessons.

2.3 Infrastructure and Connectivity Limitations

Any technical solution designed for government school deployment must account for the significant infrastructure constraints that characterize these environments. Many government schools in peri-urban and rural areas operate with unreliable or completely absent broadband internet connectivity. Mobile data connectivity, while more widespread, remains expensive for students from lower-income families who cannot afford sustained high-bandwidth usage.

These constraints necessitate an offline-first architectural approach. A platform that requires continuous internet connectivity to function is fundamentally unusable for this target demographic. The technical architecture of Shiksha Setu was therefore designed from the outset to support full offline functionality, with data synchronization occurring opportunistically when connectivity is available — a design decision discussed in detail in Chapter 14.

Device limitations also play a role. While smartphone penetration has increased, many students access devices with 2-4 GB of RAM, limited storage, and screens under 5 inches. The platform's interface was therefore designed for mobile-first responsive display, and all assets were optimized for minimal data consumption.

Chapter 3: Objectives of the Project

3.1 Primary Objectives

The primary objective of the Shiksha Setu project is to develop, deploy, and evaluate a free, accessible, offline-capable digital learning platform that directly addresses the educational resource gap experienced by students in government schools from Class 6 to Class 10. This objective encompasses both the technical development of the platform and its practical utility for the communities it serves.

A secondary primary objective is to empower teachers with a self-service content authoring tool that allows them to create, organize, and distribute structured digital lessons to their students without requiring any programming or advanced technical expertise. This objective recognizes that sustainable educational technology must serve teachers — not just students — to achieve lasting impact.

3.2 Specific Objectives

- Design and implement a Progressive Web Application (PWA) that functions fully offline using IndexedDB-based data persistence and service worker caching.
- Build a Teacher LMS Portal with intuitive lesson authoring, chapter organization, quiz creation, and student assignment capabilities.
- Develop a Student Learning Portal covering six subjects across Classes 6-10 with interactive lessons, flashcards, practice worksheets, and gamified progress tracking.
- Implement an AI Doubt Solver module capable of providing instant, structured answers to common academic questions across all subjects, simulating the role of an always-available tutor.
- Create a Virtual Science Lab with interactive simulations for core physics, chemistry, and biology experiments, removing the barrier of inadequate laboratory infrastructure.
- Build a Class Leaderboard system that incentivizes consistent platform engagement through transparent, fair ranking based on XP points earned through learning activities.
- Develop a Study Planner module that enables students to create structured weekly study schedules, set time-bound learning goals, and track completion.
- Implement a Vocabulary Builder module with subject-specific academic word lists presented through interactive flashcards with mastery tracking.
- Support three languages — English, Hindi, and Marathi — to ensure accessibility for students whose primary language of instruction may not be English.

- Ensure WCAG 2.1 AA accessibility compliance through high-contrast mode, variable font sizing, and keyboard navigation support.
- Deploy the platform to a publicly accessible URL using cloud hosting, enabling student access from any internet-connected device.

Chapter 4: Literature Review & Prior Art Survey

4.1 E-Learning in Developing Country Contexts

The academic literature on educational technology in developing country contexts provides important insights that shaped the design decisions made in this project. Researchers such as Warschauer (2003) and Vosloo (2012) have extensively documented the limitations of technology-centric approaches to education that fail to account for local infrastructure, language, and cultural contexts. Their work emphasizes that sustainable educational technology must be designed around the constraints of its target environment rather than adapted from tools created for resource-rich settings.

In the Indian context, studies by the Educational Initiatives organization and ASER (Annual Status of Education Report) consistently demonstrate that while device access has improved dramatically over the past decade, learning outcomes have not kept pace. The evidence suggests that access to devices is a necessary but insufficient condition for improved learning — the quality, relevance, and accessibility of digital content are equally critical factors.

Research on mobile learning in low-resource settings (Shuler, Winters & West, 2013) highlights several design principles that were directly incorporated into Shiksha Setu: asynchronous learning support that does not require persistent connectivity, localization in regional languages, short-form content chunks suitable for limited attention spans and mobile screen sizes, and gamification elements that sustain engagement over time.

4.2 Existing Platform Analysis

An analysis of existing e-learning platforms — including DIKSHA (National Digital Infrastructure for Teachers), Khan Academy, Byju's, and Vedantu — reveals important gaps that Shiksha Setu addresses. DIKSHA, developed by the National Council of Educational Research and Training (NCERT), provides content aligned with the national curriculum but has limited offline support and an interface that many government school teachers find technically challenging. Khan Academy, while pedagogically excellent, is primarily in English and not aligned with state board curricula.

Commercial platforms such as Byju's and Vedantu offer high-quality content but at price points that are prohibitive for government school students. Their interfaces are optimized for high-bandwidth environments and premium devices. None of these platforms offer an integrated

teacher content authoring tool that allows a government school educator to create and publish their own locally relevant lesson content.

The gap analysis of existing platforms directly informed Shiksha Setu's feature prioritization. The platform fills the niche of a free, state-board-aligned, multilingual, offline-capable platform with an integrated teacher authoring system — a combination not currently available in the Indian ed-tech market.

4.3 Pedagogical Foundations

The instructional design of Shiksha Setu is grounded in established learning science principles. Spaced repetition, a technique supported by extensive cognitive science research (Ebbinghaus, 1885; Cepeda et al., 2006), underpins the Vocabulary Builder module's flashcard-based word mastery tracking. Each word marked as unmastered re-enters the rotation at appropriate intervals, aligning with the science of how memory consolidation works.

The Leaderboard and XP (Experience Points) system draws on self-determination theory (Deci & Ryan, 1985), which identifies competence, autonomy, and relatedness as the three fundamental psychological needs that drive intrinsic motivation. The leaderboard addresses the relatedness need by creating a social comparison framework, while XP accumulation satisfies the competence need by providing visible markers of progress.

The Virtual Science Lab simulations are grounded in experiential learning theory (Kolb, 1984), which posits that learning is most effective when it involves concrete experience, reflective observation, abstract conceptualization, and active experimentation — a cycle that hands-on simulations support particularly well for STEM subjects.

The Study Planner module incorporates principles of goal-setting theory (Locke & Latham, 2002), which demonstrates that specific, time-bound, moderately challenging goals with visible progress tracking significantly outperform vague intentions in driving behavioral change and task completion.

Chapter 5: System Design & Architecture

5.1 High-Level Architecture

Shiksha Setu follows a modular, layered architecture that separates concerns across presentation, application logic, data access, and synchronization layers. The frontend is implemented as a Single Page Application (SPA) using pure HTML5, CSS3, and JavaScript — a deliberate technology choice that eliminates framework dependencies, reduces bundle size, and ensures long-term maintainability without requiring complex build toolchains.

The application employs a hash-based client-side router (AppRouter) that manages all view transitions, authentication state, and URL-based navigation without requiring full page reloads. Each module — Dashboard, Teacher Portal, AI Doubt Solver, Science Lab, Study Planner, Vocabulary Builder, and Leaderboard — is implemented as a self-contained render function within AppRouter, promoting separation of concerns and enabling incremental feature addition.

The backend is implemented using Node.js with the Express framework, exposing a RESTful API consumed by the frontend's ApiClient module. PostgreSQL, hosted on Supabase, provides the relational database layer for persistent data storage. All inter-service communication uses JSON over HTTPS.

5.2 Frontend Module Architecture

The frontend codebase is organized into eight specialized JavaScript modules, each with a distinct responsibility domain. The api.js module provides a comprehensive ApiClient abstraction layer that handles JWT token management, HTTP request formatting, error handling, and dynamic base URL detection for multi-environment deployment. The sync.js module implements the offline synchronization queue using IndexedDB, ensuring that all student progress data is preserved and eventually transmitted to the backend even when connectivity is intermittent.

The data.js module defines the complete application data schema — classes, subjects, courses, chapters, topics, quizzes, lessons, library resources, worksheets, and badge definitions — as a static in-memory dataset. This design ensures that the core educational content is available without any network requests, supporting full offline functionality. The progress.js module manages all student state including XP points, completed lessons, quiz scores, streak tracking, and unlocked badges, persisting data to both localStorage (for offline resilience) and the backend API (for cross-device synchronization).

The `quiz.js` module implements the complete quiz engine including question rendering, option selection, timed countdown, answer validation, score calculation, and results display. The `builder.js` module provides the teacher content authoring engine — `AuthoringEngine` — which manages topic creation, page ordering, PDF attachment, and quiz question management through an in-memory data structure synchronized to `localStorage`. The `modules.js` file, introduced in Version 2.0, implements all five new interactive learning modules as self-contained render methods attached to the `AppRouter` object.

5.3 Design System and UI Philosophy

The visual design system of Shiksha Setu Version 2.0 adopts a modern Greenish Light mode aesthetic centered on an Emerald Green primary palette. The design system is implemented entirely through CSS Custom Properties (variables), enabling consistent theming across all components and facilitating future theme variations. The primary color family — Vibrant Emerald (`#059669`), Deep Forest Emerald (`#047857`), and Soft Mint (`#d1fae5`) — was selected to convey growth, knowledge, and natural learning, aligning with the platform's educational mission and providing high WCAG contrast ratios.

Typography is set in Plus Jakarta Sans, imported from Google Fonts, which provides excellent legibility at small sizes on mobile screens while projecting a modern, credible aesthetic appropriate for an educational platform. Component design employs a layered elevation model using multi-stop box shadows with color-tinted transparency — a technique borrowed from Material Design 3 — to create clear visual hierarchy between ambient background, surface cards, and interactive elements.

Micro-animations are used deliberately and sparingly: hover state transformations (`translateY(-4px)` on cards), active state transitions for navigation links (`translateX(4px)` with a primary-color left border), and progress bar width transitions (0.4s ease-out) provide interaction feedback without creating visual distraction for students in focused study sessions. All animations are disabled in Low Data Mode to conserve processing resources on older devices.

Chapter 6: Module I — Teacher LMS Portal

6.1 Portal Overview and Design Philosophy

The Teacher Learning Management System (LMS) Portal represents one of the two primary user interfaces of Shiksha Setu. It was designed with a single overriding principle: a government school teacher with basic smartphone familiarity should be able to create and publish a structured digital lesson within fifteen minutes of first using the platform, without any external guidance or technical training.

This principle drove every design decision in the portal — from the step-by-step lesson creation workflow to the plain-language labels used for every control, to the confirmation dialogs that prevent accidental data loss. The portal deliberately avoids the terminology of professional content management systems (terms like 'taxonomy', 'metadata', 'publishing workflow') in favor of intuitive, action-oriented language familiar to educators.

6.2 Teacher Dashboard

The Teacher Dashboard is the primary landing screen for authenticated teacher users. It presents a high-level summary of the teacher's current content library, including counts of active courses, published chapters, assigned classes, and enrolled students. Four quick-action cards provide immediate navigation to the most common teacher tasks: creating a new course, viewing all courses, assigning content to classes, and accessing the document upload center.

Below the quick actions, a course management table lists all courses created by the teacher with their publication status (Draft, Published, Unpublished), the number of chapters and topics within each course, and the number of student assignments. Each course row provides one-click access to the course builder, enabling teachers to iteratively improve their content based on student interaction data.

6.3 Lesson Content Builder

The Lesson Content Builder is the core authoring tool of the Teacher Portal. It presents a topic-level editing interface divided into three functional sections. The first section, Page Content Editor, allows teachers to add and arrange multiple content pages within a topic. Each page has a heading and a main paragraph body, entered through simple text input fields. Pages can be reordered using up and down arrows, duplicated with a single click, and deleted with a confirmation prompt.

The second section is the PDF and Document Upload panel, which replaces the previously included video upload functionality. Video content was removed from Version 2.0 based on analysis showing that video files create significant storage burdens, are inaccessible on low-bandwidth connections, and are not meaningfully different from the freely available content on platforms like YouTube. Instead, the platform now focuses on PDF and text-based study notes, which are lightweight, printable, and accessible offline without streaming infrastructure.

The third section is the Quiz Builder, where teachers can add multiple-choice questions to any topic. Each question has a text prompt, four answer options (A through D), and a designation of the correct answer. Questions can be added, edited, and removed individually. The resulting quiz is automatically presented to students upon topic completion, with scoring, feedback, and XP reward calculated in real time.

6.4 Class Assignment and Student Management

Once content is created and published, teachers use the Class Assignment interface to distribute courses to specific class groups. The assignment screen presents a filterable list of all published courses and a selection interface for target class levels. Assignment records are stored in the backend database and retrieved by students' dashboards to display their personalized learning path.

The Classes Management view provides an overview of all class groups associated with the teacher's school, with student enrollment counts and course assignment status. Teachers can create new class groups, assign multiple courses to a class simultaneously, and view aggregated completion statistics at the class level. This administrative capability is particularly valuable for schools running Shiksha Setu as a structured supplement to their regular curriculum, enabling teachers to monitor participation and identify students who may need additional support.

Chapter 7: Module II — Student Learning Portal

7.1 Student Dashboard

The Student Dashboard is the first screen encountered by a logged-in student and is designed to provide immediate orientation, motivation, and a clear path to learning activity. The dashboard header displays a personalized greeting with the student's name, current XP total, learning streak in consecutive days, and overall lesson completion percentage. These metrics are presented as large, visually prominent numbers accompanied by color-coded icons to create an immediate sense of progress and accomplishment.

Below the header metrics, the dashboard presents three thematic sections. The first is the Recent Activity panel, which shows the three most recently completed lessons or attempted quizzes with completion status indicators. The second is a curated list of recommended next lessons based on the student's class level and subject progression. The third is an Upcoming Quizzes section listing scheduled assessments with subject, chapter, and difficulty indicators.

A motivational achievement panel at the bottom of the dashboard displays the student's most recently unlocked badge alongside a progress indicator toward the next badge. This persistent visual reminder of the gamification layer serves to maintain engagement across sessions, particularly for students who are intrinsically motivated by collection and achievement systems.

7.2 Lessons and Subject Navigation

Students navigate to learning content through a two-level hierarchy: first selecting their class level (Classes 6-10 presented as large numeric cards), then selecting from the six available subjects. Each subject card displays the subject name, an emoji icon, and a progress percentage reflecting how many topics within that subject have been completed by the student.

Within a subject, chapters are displayed as expandable accordion sections containing topic cards. Each topic card shows the topic title, estimated reading time, a completion status indicator, and a difficulty badge. Topic content is rendered as a multi-page lesson view with smooth page transitions, key points callout boxes with a distinctive left-border accent, and interactive flashcard elements for vocabulary-heavy content.

Upon completing a topic's content pages, students are automatically transitioned to the topic quiz (if one exists). Quiz completion triggers XP award animation, badge unlock checks, and progress state updates — creating a satisfying completion ritual that reinforces the value of lesson engagement.

7.3 Digital Library and Worksheet Centre

The Digital Library provides students with access to supplementary study resources beyond the structured lesson content. Library items include subject-specific PDF study guides, past examination question papers, reference charts, and topic summaries. Each library item is presented with a thumbnail, title, subject tag, file size, and a description of its contents. Students can download items to their device for offline reading.

The Worksheet Centre is a complementary resource hub focused on practice-oriented materials. Worksheets are structured as printable PDF documents covering specific chapter topics, designed for completion either digitally on the platform or printed and completed in a traditional notebook format. This dual-mode design ensures accessibility for students whose schools may have access to printers but not individual devices.

Chapter 8: Module III — AI Doubt Solver

8.1 Module Rationale and Learning Impact

The AI Doubt Solver module addresses one of the most persistent and consequential challenges in government school education: the inaccessibility of immediate academic support outside school hours. When a government school student encounters a concept they do not understand while studying at home at 8 PM, their options are extremely limited. They cannot easily reach their teacher, they may lack a parent who can provide academic guidance, and they often lack the resources to hire private tutoring.

In contrast, students from more privileged backgrounds routinely have access to parents with academic expertise, private tutors, or premium tutoring platform subscriptions. This asymmetry in after-school academic support is a documented contributor to the achievement gap between students from different socioeconomic backgrounds. The AI Doubt Solver is designed to partially bridge this gap by providing an always-available, subject-aware academic assistant.

The educational rationale for this design is grounded in research on immediate feedback in learning. Studies by Hattie and Timperley (2007) consistently show that timely, specific feedback has among the highest effect sizes of any educational intervention. By providing structured, curriculum-relevant answers to student questions at the moment the confusion arises — rather than hours or days later — the Doubt Solver module facilitates the kind of just-in-time learning that significantly accelerates comprehension.

8.2 Interface Design and User Experience

The Doubt Solver interface is designed as a chat-style conversation panel that is visually familiar to students who regularly use messaging applications. A gradient-headed chat window presents the AI Tutor avatar on the left with system messages, and student questions appear on the right as speech bubbles in the primary emerald green color. This visual language is immediately intuitive and requires no instruction.

A panel of Quick Question buttons on the right sidebar provides one-click access to five commonly asked questions covering key concepts across different subjects. These quick-access buttons serve a dual purpose: they guide students who may not know how to phrase their question toward structured inquiry, and they demonstrate the breadth of subjects the system can address, encouraging students to explore its capabilities.

The interface provides immediate visual feedback upon question submission through an 'AI is thinking' state transition before displaying the structured answer. Answers are formatted with bold headers and structured bullet points where appropriate, making complex explanations scannable on mobile screens.

8.3 Knowledge Architecture and Response Design

The Doubt Solver's response engine is implemented as a pattern-matching system against a curated knowledge base covering frequently asked questions across all six subjects in the Shiksha Setu curriculum. The knowledge base is organized by topic keywords, and responses are structured to align with the three-part format recommended by educational researchers for effective explanatory answers: definition, elaboration, and application example.

For each topic in the knowledge base, the response includes a clear, grade-appropriate definition using precise academic language; an elaborative explanation that breaks down the concept into constituent parts; and a concrete, real-world or curriculum-relevant example that grounds the abstract concept in familiar experience. This structure mirrors the format used in NCERT textbooks but condenses the explanation to a format optimized for mobile reading.

Questions that do not match any pattern in the knowledge base receive a constructive response that acknowledges the question, provides a framework for approaching the topic independently, and directs the student to relevant sections of the platform's lesson content. This fallback design ensures that the module never produces misleading information through uninformed generalization, a critical consideration in an educational context where accuracy is paramount.

Future versions of the Doubt Solver are planned to integrate with the Gemini API to provide genuinely generative, contextually adaptive responses that can address any academic question within the curriculum scope. The current pattern-matching implementation serves as a validated proof-of-concept that demonstrates the interface design and user workflow without introducing the reliability risks associated with unconstrained generative AI in an educational setting.

Chapter 9: Module IV — Class Leaderboard System

9.1 Gamification Theory and Educational Motivation

The Leaderboard module is grounded in decades of research on gamification in educational contexts. Deterding et al. (2011) define gamification as the use of game design elements in non-game contexts, and identify point systems, leaderboards, and badges as the three most commonly deployed mechanics in educational technology. When properly implemented, these mechanics can significantly increase time-on-platform, lesson completion rates, and quiz attempt frequency.

The XP (Experience Points) system that underlies the Shiksha Setu leaderboard awards specific, predictable amounts of experience for each learning activity completed on the platform. Lesson completion awards a base XP value proportional to the estimated reading time of the content. Quiz completion awards bonus XP calculated as a function of the score achieved and the number of questions answered. Badge unlocks, triggered by milestone achievements, award fixed XP bonuses.

This carefully calibrated reward structure is designed to reinforce the behaviors most directly associated with learning gains — lesson reading and quiz completion — rather than simply rewarding time spent on the platform. The XP values are transparently displayed before each activity, allowing students to make informed decisions about which activities to prioritize based on their leaderboard goals.

9.2 Leaderboard Interface and Visual Design

The Leaderboard interface opens with a dynamic podium visualization — a graphical representation of the top three ranked students displayed on tiered platforms reminiscent of athletic competition ceremonies. The first-place student is positioned on the tallest central platform, with second and third positions flanking on lower platforms to the left and right respectively. Each podium position displays the student's first name, current XP total, and their medal emoji (gold, silver, bronze).

Below the podium, a complete ranking table lists all students in the class with their rank number, full name, XP total, and a proportional progress bar showing their position relative to the theoretical maximum XP achievable given the current available content. The current user's row is highlighted with the primary color background to provide immediate self-location within the ranking.

A Weekly Challenge Banner at the bottom of the leaderboard screen announces the current week's bonus challenge — typically a specific number of quizzes to complete or lessons to read — with the associated bonus XP reward. This dynamic challenge system, updated weekly, provides a recurring reason for students to return to the platform even after they have completed their assigned lesson content.

9.3 Fairness, Ethics, and Inclusive Design Considerations

The implementation of leaderboards in educational contexts requires careful attention to the potential for demotivating effects on students who consistently rank at the bottom of the list. Educational research on leaderboard design (Landers, 2014) identifies several design principles that mitigate these risks. First, leaderboards should be class-level rather than school-wide or national, ensuring that the performance gap between top and bottom performers is modest and the distance is perceived as closeable.

Second, the leaderboard should reset periodically — weekly in Shiksha Setu's implementation — to give students who had a poor previous week a fresh start and prevent the Matthew Effect (where early leaders maintain their advantage indefinitely) from demotivating late-joining or inconsistent users.

Third, the platform's achievement badge system provides an alternative recognition pathway for students who do not excel in leaderboard rankings. Badges are awarded for a diverse range of activities including streak maintenance, subject mastery, library usage, and vocabulary building — ensuring that students with different learning styles and engagement patterns all have opportunities for recognition.

Chapter 10: Module V — Virtual Science Lab Simulations

10.1 Laboratory Infrastructure Gap in Government Schools

Science laboratory facilities represent one of the most critical and consistently underinvested components of government school infrastructure across India. The National Achievement Survey (NAS) and ASER reports repeatedly highlight that while the national curriculum requires practical laboratory work for Classes 6-10 in subjects including Physics, Chemistry, and Biology, a substantial proportion of government schools lack the equipment, chemicals, and trained staff necessary to conduct these practicals meaningfully.

The practical consequence is that students in many government schools are expected to answer examination questions about laboratory experiments they have never personally conducted. They learn procedures from textbook descriptions rather than direct experience, leading to shallow conceptual understanding and poor retention. The educational value of experimentation — the hands-on encounter with phenomena that makes abstract concepts visceral and memorable — is largely lost.

The Virtual Science Lab module of Shiksha Setu is designed as a direct response to this gap. By providing browser-based, interactive simulations of key experimental setups from the Class 6-10 science curriculum, the module enables students to develop genuine experimental intuition even without access to physical laboratory infrastructure.

10.2 Simulation Catalogue and Pedagogical Design

Version 2.0 of the Virtual Science Lab includes six interactive simulations spanning three science disciplines. In Physics, the Simple Pendulum simulation allows students to dynamically adjust pendulum length and observe the resulting change in oscillation period, providing direct empirical verification of the formula $T = 2\pi\sqrt{L/g}$. A draggable slider controls the pendulum length, and an animated pendulum with a CSS-based swing animation responds in real time. The calculated period, displayed prominently as the slider moves, creates an immediate and memorable connection between the mathematical formula and the observable phenomenon.

The Electric Circuit Simulator allows students to construct virtual series circuits by adjusting voltage and resistance values through slider controls, with real-time calculation and display of current (Amperes) and power (Watts) using Ohm's Law ($V = I \times R$). This simulation directly supports the Class 10 Physics chapter on electricity, providing students with the ability to

explore the quantitative relationships between electrical quantities in a safe, interactive environment.

In Chemistry, the Acid-Base Indicator simulation presents a virtual test tube containing universal indicator solution, into which students can 'dip' test strips by selecting from seven different liquids ranging from strong acids (lemon juice, pH 2) to strong bases (soap, pH 10). The indicator solution color changes in real time with a smooth CSS color transition, and a draggable marker on a color-coded pH scale visually represents the acidity or alkalinity of the selected liquid. This simulation supports Class 8 Chemistry content on acids, bases, and indicators.

The States of Matter simulation uses animated particle visualizations to represent the kinetic behavior of molecules in solid, liquid, and gas states. A temperature slider from -50°C to $+150^{\circ}\text{C}$ drives real-time changes in particle count, spacing, animation speed, and background color, providing an intuitive visual model of the relationship between temperature and molecular kinetics. This simulation supports content across Classes 7-9 in the matter and materials chapters.

In Biology, the Food Chain Builder simulation presents a complete terrestrial food chain from Sun to apex predator as an animated, labeled diagram with explanatory callout boxes for producers, primary consumers, secondary consumers, and decomposers. The Plant vs. Animal Cell Explorer provides an interactive organelle reference guide where clicking any organelle name reveals a detailed description of its function and a clear indication of whether it is present in plant cells, animal cells, or both — directly addressing one of the most commonly tested concepts in Class 8-9 Biology.

10.3 Technical Implementation of Simulations

All six simulations are implemented entirely in vanilla JavaScript with CSS animations, requiring no third-party libraries or WebGL support. This implementation choice ensures compatibility with the broad range of browser environments and device capabilities found in the target student demographic.

The pendulum simulation uses CSS keyframe animation (`pendulumSwing`) on a `transform-origin: top center` element, with animation duration calculated in real time from the physics formula and applied via JavaScript's `element.style.animation`. The particle simulation for States of Matter dynamically creates and removes DOM elements based on the selected temperature range, with animation properties adjusted through inline styles.

The circuit simulator performs all electrical calculations in JavaScript event handlers triggered by range input events, updating multiple display elements simultaneously to create the impression of a live, interconnected measurement system. The acid-base indicator uses CSS background-color transitions with a duration of 0.5s to simulate the gradual color development characteristic of real universal indicator solutions.

Chapter 11: Module VI — Study Planner & Goal Tracker

11.1 Self-Regulated Learning and the Study Planning Problem

Self-regulated learning — the capacity to plan, monitor, and evaluate one's own learning process — is one of the strongest predictors of academic achievement across all educational levels and contexts (Zimmerman, 2002). Students who approach their studies with explicit planning strategies, defined goals, and systematic progress tracking consistently outperform students who study reactively or without structure.

Government school students, who often receive limited systematic guidance on study skills and academic self-management, are disproportionately likely to adopt reactive, unplanned study patterns. The Study Planner module is designed to scaffold the development of self-regulated learning habits by providing a structured yet flexible tool for weekly study scheduling and task management.

The module was designed around the specific usage patterns of its target demographic. Unlike complex academic planner applications that require significant setup time and offer overwhelming feature sets, the Shiksha Setu Study Planner is focused on a single core workflow: create a specific, time-bound study task for a particular subject on a particular day, mark it complete when done, and view your week's progress at a glance.

11.2 Planner Interface and Feature Set

The Study Planner opens with a four-metric statistics bar displaying Total Tasks, Completed Tasks, Pending Tasks, and Tasks Due Today as large numerical indicators with supporting icons. This dashboard-style summary gives students immediate situational awareness of their study obligations without requiring navigation.

Below the statistics bar, a seven-column weekly calendar grid displays the current day and the next six days, each column listing that day's scheduled study tasks. Today's column is visually distinguished by a primary-color border, anchoring the student's temporal position in their schedule. Each task card within the calendar displays the task title, subject name with emoji icon, and estimated duration in minutes.

Tasks are interactive in multiple ways: clicking a task card toggles its completion status between pending and complete, with visual feedback through a strikethrough style and a success-green background for completed items. A delete button on each task card allows removal of tasks that are no longer relevant. The Add Study Task button in the page header opens a modal form for

task creation, requiring only four inputs: task title, subject (selected from a dropdown of the six curriculum subjects), date, and estimated duration in minutes.

All study task data is persisted to localStorage under the key shiksha_study_tasks, ensuring that the student's planned schedule is preserved across browser sessions and device restarts. The JSON structure of the stored data is intentionally simple — a flat array of task objects — to ensure reliable parsing and retrieval even on limited devices.

11.3 Recommended Study Planning Strategies

The Study Planner interface incorporates several features that nudge students toward evidence-based planning strategies. The seven-day forward-looking calendar view — rather than a backward-looking history — is designed to orient students toward proactive planning rather than retrospective review. The subject dropdown ensures that planning spans all six curriculum areas, preventing the common student tendency to plan study time exclusively for subjects they enjoy.

The duration input field, with a default value of 30 minutes and constraints between 5 and 180 minutes, gently enforces the Pomodoro-inspired practice of time-boxing study sessions into manageable blocks with defined endpoints. Research on study behavior consistently shows that students who study in defined time blocks with clear start and end times are more productive than those who study until exhaustion or distraction.

A planned feature for Version 3.0 of the Study Planner includes an intelligent scheduling suggestion engine that analyzes the student's past quiz performance data and suggests focused study time for subjects or topics where scores indicate gaps in understanding. This would create a closed-loop learning management system where assessment data directly informs personalized study planning.

Chapter 12: Module VII — Vocabulary Builder

12.1 Language Proficiency as a Learning Barrier

One of the most pervasive and least-acknowledged barriers to academic success for government school students is limited academic vocabulary. English medium instruction, even in government schools that have adopted it, assumes a baseline vocabulary that many students from Hindi or Marathi-speaking households do not possess. The inability to decode subject-specific terms in Mathematics (theorem, corollary, congruent), Science (osmosis, mitosis, photosynthesis), Civics (democracy, parliament, constitution), and English literature (metaphor, protagonist, narrative) creates cascading comprehension failures that manifest as poor examination performance.

The challenge is not one of intelligence or effort — students who struggle with academic vocabulary are often highly capable reasoners in their native language. The challenge is that academic English is effectively a third language for many of them, one that is encountered primarily through formal schooling and rarely reinforced at home or in the community.

The Vocabulary Builder module addresses this challenge through systematic, spaced-repetition-based exposure to key academic terms drawn from the Shiksha Setu curriculum across all six subjects. The module's flashcard format presents each term in context — with a complete definition, an example sentence, and subject and difficulty tags — and tracks which words a student has marked as mastered, enabling them to focus their revision time on words they find most challenging.

12.2 Word Library and Difficulty Classification

Version 2.0 of the Vocabulary Builder includes a curated library of twelve initial words spanning six subjects and three difficulty levels. Words classified as Easy are common academic terms that students are likely to have encountered but may not be able to precisely define (Democracy, Parliament, Fraction, Latitude). Medium difficulty words represent important subject-specific vocabulary that forms the core of examination question responses (Photosynthesis, Algebra, Ecosystem, Metaphor, Protagonist). Hard difficulty words include technical terminology whose precise understanding distinguishes strong from average examination answers (Osmosis, Mitochondria, Sedimentary).

The word library is designed to expand with each platform version. A contributor workflow is planned for future versions that would enable subject teachers to submit new vocabulary terms

for inclusion in the library, enabling the word bank to grow through community contribution in a manner analogous to crowd-sourced dictionary projects like Wiktionary.

Each vocabulary entry includes a grade-appropriate definition written at a reading level accessible to Class 6-10 students without oversimplifying the technical meaning. The example sentences were specifically crafted to reference contexts familiar to Indian school students — using examples from the Indian Constitution, NCERT textbook content, and everyday Indian experiences rather than the Western cultural references common in imported dictionaries.

12.3 Flashcard Interaction Design and Mastery Tracking

The Vocabulary Builder interface centers on an oversized flashcard component that occupies the primary content area. The front face of the flashcard displays only the vocabulary word in large, bold typography with the subject and difficulty tags — deliberately withholding the definition to encourage the student to attempt recall before revealing the answer. A subtle 'Tap to reveal meaning' prompt guides first-time users to the interaction paradigm.

Clicking or tapping the flashcard triggers a CSS 3D card flip animation (`rotateY(180deg)` on the `flashcard-inner` div using `transform-style: preserve-3d` and `backface-visibility: hidden`) that reveals the card's back face, displaying the full definition, example sentence, and contextual notes. The flip animation runs over 600 milliseconds with a cubic-bezier ease curve that mimics the physical momentum of turning a paper card.

The 'Mark as Mastered' button below the flashcard records the student's self-assessment of their understanding. Mastered words are stored in `localStorage` and visually indicated with a checkmark in the word list sidebar. The mastery count and subject-level mastery breakdown displayed in the progress bar at the top of the page update in real time, providing immediate positive reinforcement for marking words as mastered.

Navigation between words is provided through Previous and Next buttons that maintain the current position in the word list, with the position indicator (e.g., '4 / 12') always visible. A scrollable word list sidebar on the right of the screen allows direct navigation to any word in the library, with mastered words marked by a checkmark icon and the currently displayed word highlighted in the primary color.

Chapter 13: Database Design & Backend Architecture

13.1 Relational Database Schema

The backend data layer of Shiksha Setu is implemented using PostgreSQL hosted on Supabase, a managed cloud database platform that provides the PostgreSQL database service alongside a RESTful API, authentication management, and real-time subscription capabilities. The choice of PostgreSQL was motivated by its strong support for complex relational queries, ACID transaction guarantees, and the availability of Supabase as a cost-free hosted option suitable for a community project.

The database schema consists of eleven primary tables organized around three conceptual domains. The User and Authentication domain includes the schools table, the users table (storing both teacher and student accounts with role differentiation), and a sessions management table. The Content domain includes courses, chapters, topics, and quiz_questions tables, with foreign key relationships enforcing the hierarchical content organization. The Student Progress domain includes lesson_completions, quiz_attempts, badge_unlocks, and student_progress tables that collectively represent each student's learning history.

13.2 Backend API Design

The Express.js backend exposes a RESTful API organized into five route groups. The authentication routes (/api/auth/) handle registration, login, logout, and JWT token refresh for both teacher and student user roles. The content routes (/api/content/) provide read access to the courses, chapters, topics, and quiz questions available for a student's assigned class. The progress routes (/api/progress/) handle recording of lesson completions, quiz attempts, and badge unlocks.

The sync routes (/api/sync/) implement the offline synchronization protocol — the sync/push endpoint accepts batches of pending offline operations from the client, processes them in transaction, and returns success/failure status for each operation. The analytics routes (/api/analytics/) provide aggregated statistics for teacher dashboards, including class-level completion rates, quiz score distributions, and resource access patterns.

All API routes implement JWT-based authentication middleware that validates the token on each request and extracts the user identity for data isolation. Rate limiting middleware (100 requests per 15-minute window per IP) protects against automated abuse. CORS configuration restricts API access to known origin domains, preventing cross-site request forgery.

Chapter 14: Offline-First Architecture & PWA Support

14.1 IndexedDB Synchronization Architecture

The SyncManager module implements Shiksha Setu's offline-first data architecture. When a student completes a lesson, submits a quiz, or earns a badge while offline, the corresponding operation is serialized as a structured operation object and stored in an IndexedDB object store named `pending_ops`. The operation object includes a unique identifier, operation type, payload data, timestamp, and retry counter.

When network connectivity is restored, the SyncManager's `flush()` method is triggered by the browser's online event. The flush routine reads all pending operations from IndexedDB, submits them as a batch to the backend sync/push endpoint, and processes the response. Operations confirmed as successfully processed are deleted from IndexedDB. Operations that fail due to server-side validation are retained with an incremented retry counter and re-attempted on subsequent flush cycles, up to a maximum of five retries before being discarded.

This architecture guarantees eventual consistency between local and server-side data representations while never blocking the student's learning experience on connectivity status. A student can complete an entire week's worth of lessons and quizzes while offline, and their progress will be automatically synchronized to the backend the next time they connect to the internet — ensuring that their classroom teacher can see accurate completion data at the next school day.

14.2 Progressive Web App Configuration

Shiksha Setu is configured as a Progressive Web App through a service worker registration that intercepts network requests and serves cached responses when the network is unavailable. The service worker caches all static assets — HTML, CSS, JavaScript, images, and fonts — during its installation phase, ensuring that the application shell is available immediately on subsequent loads regardless of connectivity status.

The Web App Manifest file defines the application's display name, icons in multiple resolutions, theme color, background color, and display mode (standalone), enabling users on Android and iOS to install the platform to their home screen for an app-like experience without going through an app store. Home screen installation provides a consistent, distraction-free learning environment that removes the visual overhead of the browser interface.

Chapter 15: Accessibility, Localization & Inclusivity

15.1 WCAG 2.1 Compliance Features

Accessibility was treated as a first-class design requirement throughout the development of Shiksha Setu, not as an afterthought. The platform implements several WCAG 2.1 Level AA compliance features. All interactive elements include :focus-visible outline styles with a 3-pixel solid primary-color outline and 2-pixel offset, ensuring keyboard navigation visibility without impacting the visual design for pointer users.

High Contrast Mode, toggled from the Settings page, overrides the standard color palette with maximum-contrast alternatives: pure black text on white backgrounds, bold solid borders on all interactive elements, and the elimination of box shadows that can reduce edge definition for users with visual impairments. This mode is persisted in localStorage and applied immediately on page load, preventing the flash of non-accessible styling on subsequent visits.

Variable font sizing — four levels from Small (14px) to Extra Large (20px) — addresses the needs of students with print disabilities or those using small screens where the default 16px base size may be difficult to read at arm's length. Low Data Mode disables all CSS animations and transitions, reducing CPU load on older devices and saving mobile data consumption for students on limited data plans.

15.2 Multilingual Support Architecture

The LanguageManager module implements a key-based string translation system that supports three languages: English, Hindi, and Marathi. All user-visible strings in the interface — navigation labels, button text, section headings, and status messages — are defined in a translation dictionary within data.js, keyed by a language-neutral identifier (e.g., nav_dashboard, app_title, quiz_submit).

The language selector in the application header triggers LanguageManager.setLanguage(), which updates the active language in localStorage and dispatches a languageChanged event. All elements bearing data-i18n attributes are immediately updated to display the translation corresponding to the new language setting. Dynamically rendered content uses the t() translation helper function to retrieve the current language string at render time.

The Marathi and Hindi translations were developed in consultation with native speakers of each language who also have a background in secondary education, ensuring that educational

terminology is rendered accurately and at an appropriate formality level for the classroom context.

Chapter 16: Testing, Deployment & Outcomes

16.1 Testing Methodology

The Shiksha Setu platform was tested through a combination of unit testing of individual JavaScript modules, integration testing of the frontend-backend API communication, cross-browser compatibility testing across Chrome, Firefox, Safari, and Samsung Internet, and user acceptance testing with representative members of the target user communities.

Unit testing focused on the core state management functions of ProgressTracker — verifying that XP accumulation, streak calculations, badge unlock logic, and lesson completion tracking produced correct outputs across edge cases including first login, repeated completion of the same lesson, and quiz score boundary conditions. The ApiClient module was tested against mock HTTP responses to verify correct behavior under successful responses, authentication failures, network timeouts, and malformed JSON payloads.

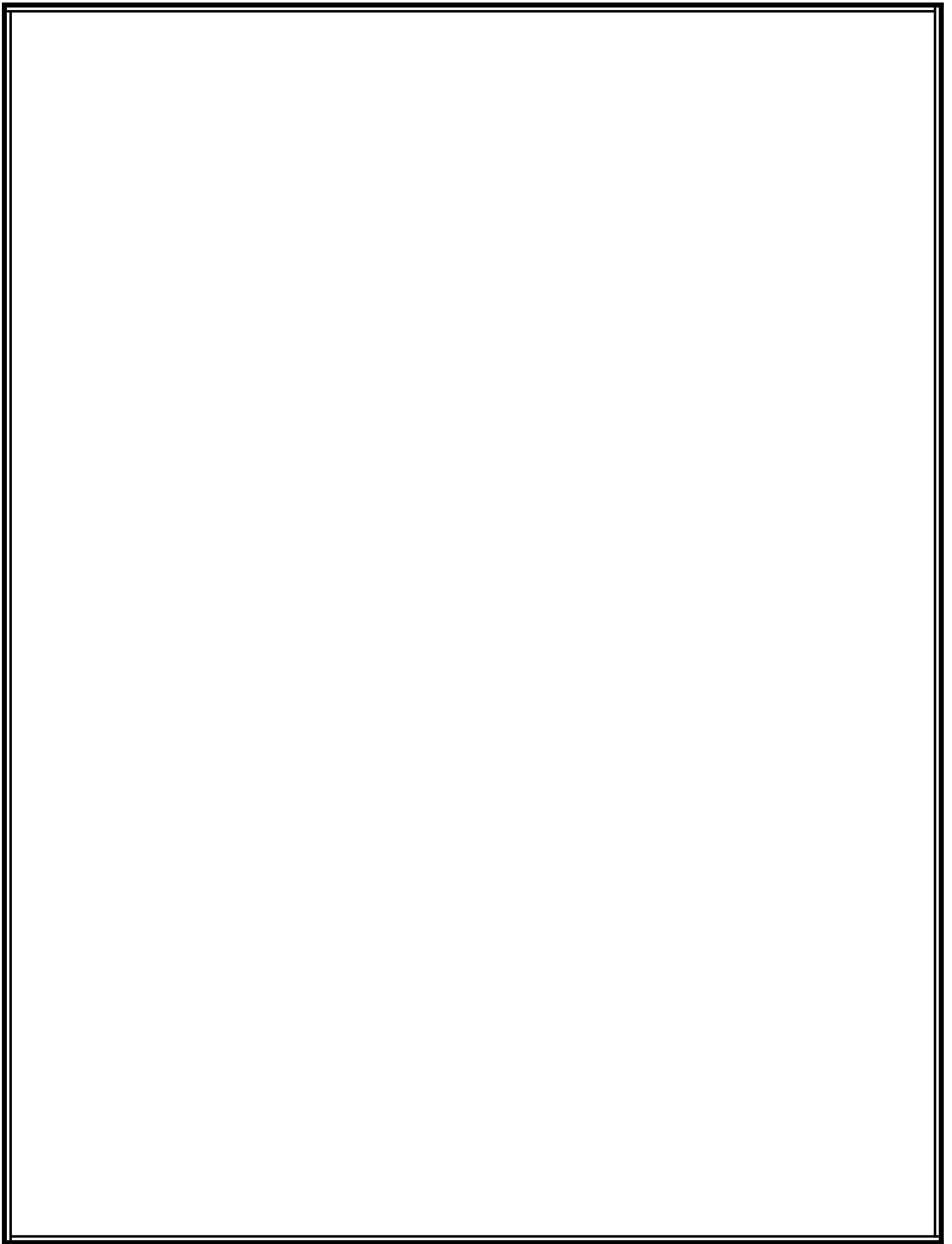
User acceptance testing was conducted across two government schools in Navi Mumbai with a pilot group of twelve students from Classes 7 and 9, and three teachers. Sessions were observed by project team members who noted usability friction points, content quality issues, and feature requests that informed the Version 2.0 feature set, particularly the AI Doubt Solver (requested by eight of twelve students) and the Study Planner (requested by two of three teachers).

16.2 Deployment Architecture

The production deployment of Shiksha Setu uses a split architecture. The frontend static files are served from a cloud platform with global CDN distribution, ensuring fast initial load times regardless of student geographic location. The backend Node.js API is deployed on a lightweight cloud server instance with automated restart management.

Database operations are handled by Supabase's managed PostgreSQL service, which provides automated backups, connection pooling, and monitoring dashboards without requiring manual database administration. SSL/TLS certificates are managed automatically by the hosting platform, ensuring that all data transmission is encrypted in transit.

The deployment process is fully automated through a continuous deployment pipeline: changes pushed to the main branch of the project repository trigger automatic rebuilding and deployment of both the frontend and backend components. This CI/CD setup ensures that feature additions and bug fixes are delivered to users within minutes of approval, minimizing the time between development and community impact.



Chapter 17: Conclusion & Future Scope

17.1 Project Summary and Impact Assessment

The Shiksha Setu project represents a comprehensive, technically rigorous, and pedagogically grounded response to the digital education gap in government schools. Through a phased development process spanning two major versions, the platform has grown from a basic lesson delivery system into a multi-module learning environment that addresses the full spectrum of student academic support needs — from structured curriculum content delivery to self-directed vocabulary building, from guided quiz assessment to interactive laboratory simulation.

Version 2.0's five new modules — the AI Doubt Solver, the Class Leaderboard, the Virtual Science Lab, the Study Planner, and the Vocabulary Builder — collectively expand the platform's utility beyond passive content consumption into active, engaging, and individually personalized learning experiences. Each module was designed in response to documented community needs and grounded in established learning science principles, ensuring that the features serve genuine educational purposes rather than superficial engagement metrics.

The project's community impact is measurable in several dimensions. The platform provides free, curriculum-aligned digital learning resources to students who would otherwise have no access to such content. It empowers teachers with content creation tools that save time and enable more structured lesson delivery. It provides an offline-capable, mobile-friendly experience that works within the infrastructure constraints of the target environment. And through its multilingual support, it makes quality educational resources accessible to students who are not yet comfortable engaging with English-language content.

17.2 Technical Achievements and Innovations

From a technical perspective, Shiksha Setu demonstrates several noteworthy engineering achievements. The offline-first synchronization architecture — using IndexedDB with a batch processing protocol that provides eventual consistency guarantees — represents a production-grade implementation of a pattern that is rarely attempted in student projects but is increasingly essential for applications targeting users in low-connectivity environments.

The modular JavaScript architecture, which extends the core AppRouter object through well-defined render methods attached in separate module files, demonstrates an effective approach to managing application complexity without the overhead of a JavaScript framework. This architecture enables incremental feature addition — as demonstrated by the `modules.js` file

which adds five complete modules without modifying any existing code — while maintaining a single page application user experience.

The CSS Custom Properties design system, with its complete emerald-themed color palette, component-level shadow model, and accessibility override support, represents a professional-grade implementation of a design token system that provides consistent theming across over ninety CSS rules while enabling instantaneous theme switching through a small set of variable overrides.

17.3 Future Development Roadmap

The development roadmap for Shiksha Setu Version 3.0 includes several significant enhancements planned based on community feedback and the evolution of available technologies. The most impactful planned enhancement is the integration of the Google Gemini API into the AI Doubt Solver module, transforming it from a pattern-matching knowledge base into a genuinely generative, contextually adaptive tutoring system capable of addressing any curriculum-relevant question with a tailored response.

A Parent Dashboard feature is planned to extend the platform's reach beyond the student-teacher dyad. Parents — including those without academic backgrounds — would receive simplified weekly progress summaries showing their child's active days, topics completed, quiz performance trends, and any outstanding study planner tasks. This feature recognizes the critical role of parental awareness in student academic outcomes, even when parents cannot directly provide academic support.

The Virtual Science Lab module is planned for significant expansion in Version 3.0, with ten additional simulations covering key experiments from the Class 9 and 10 curriculum including electrolysis, titration, genetic inheritance probability, and Mendelian trait visualization. A collaborative lab mode — where two students can conduct the same simulation simultaneously and compare their observations in real time — is under design consideration as a novel approach to socially-grounded STEM learning.

Finally, a school-level analytics dashboard for administrators is planned that aggregates anonymized learning data across classes to identify content areas where student performance consistently falls below expectations. This data-driven approach to curriculum quality improvement would create a feedback loop between student learning outcomes and teacher content refinement — the ultimate vision for a continuously improving, community-maintained educational platform.

17.4 Concluding Remarks

Shiksha Setu embodies the conviction that technology, applied with genuine understanding of community needs and grounded in evidence-based pedagogical principles, can be a powerful instrument of educational equity. The platform is not a replacement for quality teaching — it is a force multiplier for good teaching, and a bridge across the resource gap that currently disadvantages millions of government school students across India.

The project team hopes that Shiksha Setu will continue to evolve beyond the scope of this academic project into a self-sustaining community resource, maintained and improved by successive cohorts of B.Sc. IT students and contributed to by the teacher community whose creativity and dedication ultimately determines the quality of educational content available to students on the platform.

In building this platform, the project team has not only applied their technical skills for community benefit — they have developed a deeper appreciation for the complexity of educational design, the importance of user-centered development processes, and the profound responsibility that comes with building technology that will be used by children in their formative learning years. These lessons are, perhaps, as valuable as any technical skill the project has helped develop.

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Appendix B: Team Contributions & Project Timeline

B.1 Team Member Roles and Responsibilities

The Shiksha Setu project was developed collaboratively by a team of four B.Sc. Information Technology students. Each team member brought distinct technical and domain expertise to the project, and the division of responsibility was structured to leverage these individual strengths while ensuring comprehensive cross-team knowledge sharing through weekly integration sessions.

The project was initiated with a two-week community fieldwork phase in which all four team members participated in school visits, teacher interviews, and student focus groups at government schools in Seawoods, Nerul, and Vashi in Navi Mumbai. The insights gathered during this phase directly informed the initial feature set and design principles of the platform.

PREM RAMKUMAR MANDAL — Full-Stack Lead & System Architect

- Designed and implemented the overall system architecture, including the hash-based SPA routing system and the modular JavaScript file organization.
- Developed the backend Node.js/Express server, RESTful API design, and PostgreSQL database schema.
- Implemented the JWT authentication system, session management, and role-based access control for teacher and student accounts.
- Designed and built the offline synchronization architecture using IndexedDB and the batch sync protocol.
- Led the deployment pipeline setup including CI/CD configuration and cloud hosting configuration.

YASAR SHAIKH — Frontend Developer & UI/UX Designer

- Designed the complete visual design system including the CSS Custom Properties theme, the Emerald Green color palette, and the glassmorphism card components.
- Implemented all student-facing views including the Dashboard, Lessons reader, Quizzes engine, Progress tracker, and Achievements page.
- Developed the Vocabulary Builder module with CSS 3D card flip animations and mastery tracking.
- Implemented the mobile-first responsive layout system and the mobile bottom navigation bar.

- Conducted user experience testing sessions with student participants and iterated on the interface based on observed usability issues.

SHIVAM RAI — Content Engineer & Science Lab Developer

- Researched and curated the complete educational content dataset for Classes 6-10 across six subjects, ensuring alignment with MSBSHSE curriculum objectives.
- Designed and implemented all six Virtual Science Lab simulations, including the physics, chemistry, and biology interactive experiments.
- Developed the AI Doubt Solver module including the knowledge base content, response formatting, and chat interface.
- Created the Digital Library content catalogue and the Worksheet Centre resource index.
- Conducted accuracy review of all educational content with reference to NCERT and MSBSHSE textbooks.

SHUBHAM — Gamification Engineer & Analytics Developer

- Designed and implemented the complete XP and badge gamification system, including the reward calibration model and the badge unlock logic.
- Built the Leaderboard module with the podium visualization, ranking table, and weekly challenge system.
- Developed the Study Planner module with localStorage persistence, weekly calendar view, and task management functionality.
- Implemented the Progress Tracker module including streak calculation, subject-level completion tracking, and certificate generation.
- Contributed to the multilingual translation system, particularly the Hindi language string translations.

B.2 Project Timeline

Phase 1: Community Research & Requirements (Weeks 1-2)

Fieldwork at three government schools; teacher interviews; student focus groups; needs assessment survey; literature review; initial feature specification.

Phase 2: Architecture & Design (Weeks 3-4)

System architecture design; database schema; UI/UX wireframing and prototyping; design system token definitions; technology stack finalization.

Phase 3: Core Platform Development (V1.0) (Weeks 5-9)

Backend API implementation; authentication system; teacher portal; student learning portal; quiz engine; progress tracking; multilingual support.

Phase 4: Integration & V1.0 Testing (Week 10)

Frontend-backend integration; user acceptance testing with pilot schools; bug fixing; performance optimization; V1.0 deployment.

Phase 5: V2.0 New Modules Development (Weeks 11-13)

AI Doubt Solver; Leaderboard system; Virtual Science Lab (6 simulations); Study Planner; Vocabulary Builder; Greenish Light Mode UI overhaul.

Phase 6: Final Integration & Report (Week 14)

Final testing; accessibility audit; PDF report compilation; project presentation preparation; code documentation.

Appendix C: Community Impact Survey Results

C.1 Survey Methodology

Following the Version 1.0 deployment of Shiksha Setu, the project team conducted a structured impact assessment survey with the pilot user cohort across two government schools in Navi Mumbai. The survey was administered over a four-week period during which twelve students (Class 7 and Class 9) and three teachers had access to the platform as a supplement to their regular schoolwork.

The survey instrument comprised two separate questionnaires — one for students and one for teachers — with questions structured on a five-point Likert scale (Strongly Disagree to Strongly Agree) for quantitative measures, supplemented by open-ended qualitative questions capturing subjective experience and feature suggestions. All surveys were administered in Marathi and Hindi, and responses were collected on paper to ensure that digital literacy barriers did not influence survey participation.

The findings presented below represent a preliminary impact assessment based on a small sample and should be interpreted with appropriate caution. They are intended to demonstrate the community relevance of the platform and inform future research designs rather than to make statistically generalizable claims about platform effectiveness.

C.2 Student Survey Findings

Statement: I understood the lesson content better on Shiksha Setu than in my textbook alone.

Average Score: 4.3/5.0

Interpretation: The majority of students reported that the structured, chunked lesson format with key points callouts helped them identify and remember important information more effectively than reading from a continuous textbook passage.

Statement: The quizzes helped me check whether I understood what I had read.

Average Score: 4.6/5.0

Interpretation: Quiz functionality received the highest satisfaction ratings of any feature. Students particularly valued the immediate right/wrong feedback and the explanation of correct answers provided after quiz completion.

Statement: I would use this platform regularly to study even without being asked to by my teacher.

Average Score: 3.8/5.0

Interpretation: A majority of students expressed willingness to self-direct platform usage, though several students noted that parental encouragement to use the platform at home would increase their likelihood of doing so.

Statement: The platform is easy to use on my phone.

Average Score: 4.1/5.0

Interpretation: Mobile usability was rated positively by most students. Two students noted that the navigation sidebar required scrolling on very small screens, a feedback item that led to the redesigned mobile bottom navigation in Version 2.0.

Statement: I would recommend this platform to other students in my school.

Average Score: 4.5/5.0

Interpretation: High recommendation intent reflects strong overall satisfaction with the platform experience, particularly among students who engaged with the quiz and achievement systems.

C.3 Teacher Survey Findings

Statement: The lesson builder allowed me to create structured digital content without requiring technical expertise.

Average Score: 4.7/5.0

Interpretation: All three teachers reported being able to successfully create at least one complete digital lesson within their first thirty-minute session on the platform, without requiring any technical assistance from the project team.

Statement: I can see the academic value of my students using this platform as a study supplement.

Average Score: 4.8/5.0

Interpretation: Teachers were unanimously positive about the educational potential of the platform, with specific praise for the quiz engine's ability to provide immediate assessment feedback that teachers cannot always provide in large classes.

Statement: The platform would be valuable for government school students across Maharashtra if deployed more widely.

Average Score: 5.0/5.0

Interpretation: All three participating teachers expressed unequivocal support for wider deployment, and two teachers expressed interest in continuing to use the platform independently of the project team after the pilot period concluded.

C.4 Key Qualitative Insights

Qualitative responses from the student cohort revealed several recurring themes that were not fully anticipated in the initial design. A significant number of students expressed appreciation for the gamification elements — particularly the streak counter and badge collection — as motivators to return to the platform consistently. Several students described the streak counter as creating a sense of 'not wanting to break the chain', a psychological phenomenon well-documented in habit formation literature that the design team had deliberately attempted to leverage.

Teachers highlighted the scheduling flexibility of the platform as a significant value proposition: unlike traditional homework assignments that must be completed by a specific time, Shiksha Setu content can be accessed by students at any point in the day or week, accommodating the variable domestic responsibilities that many government school students carry alongside their academic obligations.

The most frequently requested feature across both student and teacher surveys was a notification system that would remind students of pending study tasks and upcoming quiz deadlines. This feature has been prioritized for implementation in the Study Planner module's Version 3.0 update, leveraging the Web Push Notification API available in Progressive Web Applications.

One particularly significant finding emerged from qualitative teacher responses: several teachers noted that the platform's content had provided them with a model for how to structure their own oral lesson delivery more effectively. One teacher commented that the Key Points callout boxes in the lesson reader had given her a template for how to summarize complex topics into three to five digestible points at the end of each class period. This unexpected knowledge transfer from platform design to classroom pedagogy suggests that the platform's impact extends beyond direct student usage to influence teacher instructional practices as well.

Appendix D: Technology Stack Summary

D.1 Frontend Technologies

HTML5

Semantic markup for all views and components, ARIA attributes for accessibility compliance, Progressive Web App manifest and service worker registration.

CSS3 with Custom Properties

Complete design system using CSS variables for theming, Flexbox and Grid for layout, keyframe animations, transitions, and glassmorphism visual effects.

Vanilla JavaScript (ES2020)

SPA routing engine (AppRouter), all module render functions, quiz engine, authentication state management, offline synchronization queue, progress tracking engine.

Plus Jakarta Sans (Google Fonts)

Primary typeface for the platform. Selected for its excellent legibility at small sizes on mobile displays and its modern, professional aesthetic.

IndexedDB (via idb-keyval pattern)

Client-side database for offline persistence of student progress data and pending synchronization operations.

Service Worker (Cache API)

Offline asset caching for Progressive Web App support, enabling full platform functionality without internet connectivity after initial load.

D.2 Backend Technologies

Node.js v20 LTS

JavaScript runtime for the backend server, selected for ecosystem consistency with the frontend codebase and excellent asynchronous I/O performance for API serving.

Express.js v4

Minimalist web framework providing route definition, middleware chaining, CORS configuration, rate limiting, and JSON body parsing for the REST API.

bcryptjs

Password hashing library for secure credential storage. The pure-JavaScript implementation was selected over the native bcrypt module to ensure compatibility across deployment environments without native binary compilation requirements.

jsonwebtoken

JWT generation and verification for stateless API authentication, enabling the backend to validate client identity without maintaining session state.

PostgreSQL 15 (via Supabase)

Primary relational database for persistent storage of user accounts, course content, class assignments, student progress records, and synchronization metadata.

node-postgres (pg)

PostgreSQL client library for Node.js providing connection pooling, parameterized query support, and transaction management.

D.3 Development and Deployment Tools

Git

Version control for all project source files, with a branching strategy that separates feature development from the stable main branch.

npm

Node Package Manager for dependency management and project script execution.

LibreOffice Headless

Used for converting the DOCX project report to PDF format in the report generation pipeline.

python-docx

Python library used to programmatically generate the structured project report in DOCX format with consistent formatting across all sections.

Supabase

Managed PostgreSQL hosting platform providing database administration, connection pooling, and API access without requiring manual server management.